



Friction modeling, identification, and compensation based on friction hysteresis and Dahl resonance



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ABSTRACT

This paper studies effects of friction on control systems and utilizes the observed frictional behavior to develop a parameter identification method for a friction model using frequency domain measurements. Friction exists in a wide range of drive systems due to physical contacts in bearing elements, transmissions, or motion guides. Friction in a control system can deteriorate performance by causing limit cycles or stick-slip, as well as larger tracking errors. Friction compensation can help to reduce following errors, but requires physical understanding and a reliable model of friction in both the gross- and the pre-sliding regimes. In this paper, we adopt the Generalized Maxwell-Slip (GMS) model and develop a frequency-domain method to identify the model parameters based on the frictional resonances, which occur due to the elastic behavior of friction at small amplitudes. With the experimentally identified parameters, the friction model is utilized to compensate the friction effects in a motion control system. The resulting system performance of a compensated and uncompensated control system is then compared in both the frequency and time domains to demonstrate the Dahl resonance identification method for a GMS model.

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1. Introduction

Friction behavior can be divided into two regimes: gross-sliding and pre-sliding [1]. In the gross-sliding or simply sliding regime, friction is a function of the relative velocity of two sliding objects and is well established as the Stribeck curve containing the elements of static, Coulomb, and viscous friction [2]. Friction in the pre-sliding regime, on the other hand, is not directly a function of velocity but depends upon the history of displacement of objects in contact, acting as a nonlinear hysteretic spring [3].

In this paper, we experimentally explore friction behavior in the frequency domain, and use that perspective to identify parameters of the Generalized Maxwell-Slip (GMS) friction model [4]. A frequency domain identification was introduced by Hensen et al. [5] for an early dynamic friction model, called the LuGre model [6]. The identification approach developed in this paper extends the frequency-domain view to extract the multiple varying stiffnesses of the pre-sliding friction in the GMS model based on the frictional

resonance, which is a frequency-domain reflection of the hysteretic nonlinear behavior of the pre-sliding friction. We experimentally validate the fidelity of our identification method in both the frequency and the time domains by comparing system performance with and without a model-based friction compensator.

Prior to an in-depth discussion of empirical friction models, friction model parameter identification, and friction compensation, we present a case study using a servomotor to help physically understand friction behaviors.

2. Case study: Motor spin free response

2.1. Experimental setup

Fig. 1 shows a friction experimental testbed including a servomotor and a high-resolution encoder. We use a slotless and brushless servomotor, BMS 60, by Aerotech so as to minimize unwanted nonlinear effects including brush friction and cogging. A high performance rotary encoder implemented at the back of the motor keeps track of the position displacement with an interpolated resolution of 10^6 counts per revolution. The position information goes through an FPGA encoder counter into a controller which is implemented on a National Instruments (NI) PXI 8110 real-time control-

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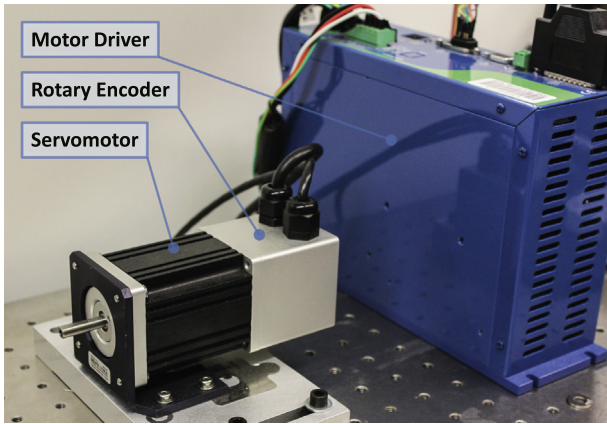


Fig. 1. Experimental setup including an Aerotech servomotor (BMS 60), encoder, and motor driver (Soloist Hpe 10).

ler connecting to the host computer via ethernet. We use this experimental setup for the case study, and for the rest of the experiments in this paper as well.

2.2. Free response of motor in rotation

Two simple experiments demonstrate the effects of bearing friction. In the first test, we manually spin the motor shaft by hand with the motor coils disconnected (open circuit) while reading motor position and velocity. The resultant time responses are shown in Fig. 2(a) and (b). The position response is measured at a sampling rate of 2 kHz, and the velocity is calculated in the servo controller by the backward difference of the measured position [7]. For the second test, the motor is excited by a current pulse with the amplitude of 0.05 A and a duration of 70 ms. Fig. 2(c) and (d) show the resultant responses of the motor driven open-loop test. The comparison between the manual and the open-loop tests indicates that the resulting oscillatory behavior is not caused by any electrical components of the system, but by friction from the direct physical contact of the mechanical parts, which is the bearing in this setup.

The spins illustrated in Fig. 2(a) and (c) are roughly 0.55 and 1.17 rad in amplitude with peak velocities of 7.6 and 15.4 rad/s respectively. For both spins, the friction of the motor ball bearing acts as a near-constant drag torque during continuous rotation. However, for the region where the velocity becomes low, the friction shows elastic behavior, which causes oscillations as seen in the magnified views. Note that the frequency of the oscillation is on the order of tens of Hertz, thereby implying that it cannot be due to mechanical eigenfrequencies such as due to motor flexibility.

The results of these tests show key aspects of the friction in both the gross- and pre-sliding regimes. In the gross-sliding regime, the friction opposes the relative velocity with nearly constant torque. In the pre-sliding regime as the rotor comes to rest, friction acts as an elastic element which resonates with the rotor inertia. Note that the oscillation period decreases as the shaft comes to the stop, as illustrated in Fig. 2(b), which indicates that the stiffness of the friction elasticity increases for smaller displacements. This observation agrees with the nonlinear hysteretic behavior of pre-sliding friction as reported in many references including [1,3] and with the GMS model adopted in this paper. Also note that the open-loop test shows a bit more lightly-damped response, but given that the two tests were conducted several months apart at different shaft positions with different environmental conditions including temperature and humidity, this discrepancy seems insignificant.

2.3. Paper outline

Both regimes should be taken into account in constructing a friction model and compensating the friction, especially the pre-sliding regime for high precision applications, since a positioning system enters this regime frequently and can be dominated by the pre-sliding friction. In Section 3, we discuss the classic and modern empirical friction models, which are efforts to mathematically represent the friction in both regimes. In this paper we use the Generalized Maxwell-Slip (GMS) model which has advantages of high fidelity and relatively simple implementation for real-time control purposes [4,8]. We develop and discuss a frequency-domain method to identify parameters of this model in Section 4. A model-based friction compensation technique is studied and the resultant performance is shown in both the frequency and the time domains in Section 5. Section 6 concludes the paper with final comments.

3. Prior art

In this section, we review representative aspects of friction together with several empirical friction models which are introduced in the literature and relevant to our study in this paper.

3.1. Classic friction model

The simplest model of the friction drag force is $F = \mu N$ where μ is the friction coefficient and N is the normal force. This model is often referred to as Coulomb friction. When the model is augmented with static friction and a linear viscous drag term, the force-velocity graph appears as plotted in Fig. 3(a). A model with continuous velocity dependency in the sliding regime was developed by Stribeck [9] in the form of

$$F(v) = s(v) + \sigma v = \text{sgn}(v) \left(F_c + (F_s - F_c) \exp \left| \frac{v}{V_s} \right|^\delta \right) + \sigma v,$$

and is illustrated in Fig. 3(b). This model includes viscous friction, where F_c , F_s , V_s , δ and σ represent the Coulomb friction, static friction, Stribeck velocity, shape factor, and viscous friction coefficient respectively. Note that in the Stribeck curve, $F(v)$ has both a velocity weakening curve, $s(v)$, and a velocity strengthening curve, σv [10].

The Stribeck curve is often called static since the curve equation is only a function of the relative velocity of the sliding objects, and can be experimentally obtained by measuring friction force over a range of constant velocities. The model's biggest drawback is the discontinuity at zero velocity, which is the pre-sliding regime. This limitation means that the Stribeck model cannot encompass dynamical phenomena including friction lag [11], rate-dependent breakaway force [12,13], and hysteresis with non-local memory [3,14,15].

3.2. More recent empirical friction models

There have been many efforts to develop friction models which effectively cover the sliding and pre-sliding friction regimes, and a variety of empirical models have been introduced in the literature attempting to represent the hysteretic elastic behavior of the pre-sliding regime depicted in Fig. 3(d) as well as the classic aspects of sliding friction.

Dahl [16,17] introduced a model describing the pre-sliding regime friction, in 1968, based on the speculation that the relationship of friction force and displacement resembles the stress-strain behavior of ductile materials. Introducing the LuGre model in 1995, Canudas de Wit et al. [6] constructed a state evolving equation

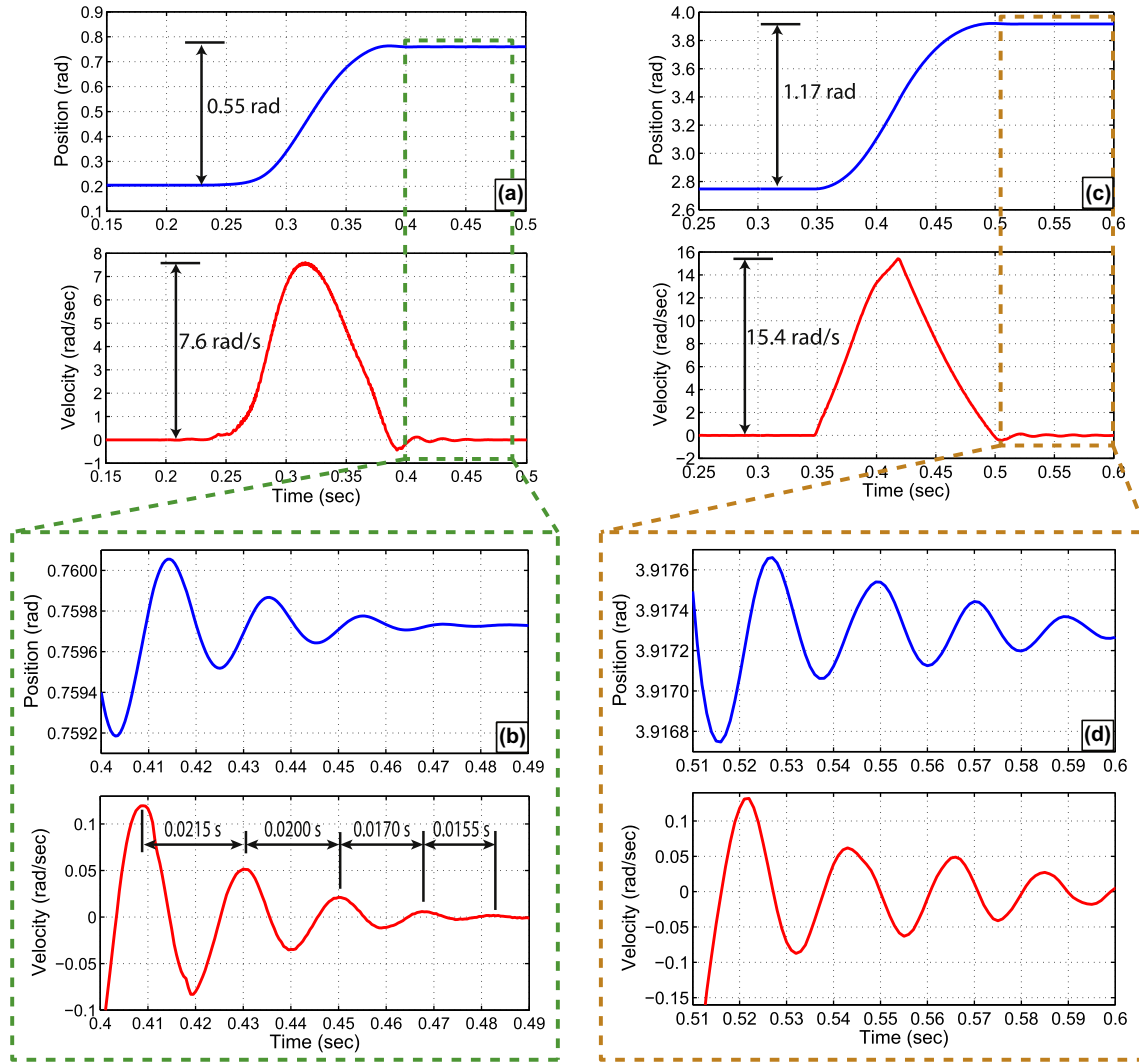


Fig. 2. Responses of a motor shaft. (a) Manual test: angular position and velocity when spun by hand. (b) Manual test: magnified view of oscillation in angular position and velocity due to friction as the rotor comes to rest. (c) Motor driven open-loop test: angular position and velocity when excited by the phase current. (d) Motor driven open-loop test: Magnified view of the oscillation showing the similar behavior as the manual test.

with the pre-sliding displacement as the internal state to model the friction in both regimes with a smooth transition from the pre-sliding to sliding regime.

In order to describe the non-local memory characteristic, observed in the friction hysteresis when there are multiple velocity reversals in the displacement as illustrated in Fig. 3(c) and (d), the LuGre model was modified to develop the Leuven model by Swevers et al. in 2000 [15]. This model adds a hysteresis function to represent the non-local memory, and the hysteresis function is constructed with the Maxwell-Slip (MS) model [18] of parallel-connected elementary blocks as shown in Fig. 4(a), each of which has the elastic behavior k_i of Fig. 4(b) with the Coulomb slip law of force limit $\alpha_i F_c$.

Instead of the constant Coulomb slip law, Al-bender et al. [8] imposed sliding dynamics as a weighted Stribeck velocity-weakening curve $\alpha_i s(v)$ on the slipping phase of each Maxwell-Slip block and constructed a friction model, called the Generalized Maxwell-Slip (GMS) model. The behavior of each GMS block is illustrated in Fig. 4(c). In the GMS model, two states of either stick or slip determine the characteristic behavior of each elementary block with the rate-state law, which are mathematically represented as

$$\begin{aligned} \frac{dF_i}{dt} &= k_i v, \quad \text{if stick} \\ \frac{dF_i}{dt} &= \text{sgn}(v) C \left(\alpha_i - \frac{F_i}{s(v)} \right), \quad \text{if slip} \\ F_f &= \sum_{i=1}^N F_i(t) + \sigma v(t). \end{aligned}$$

Here C indicates an attraction factor determining how fast the slipping dynamics follows the velocity weakening curve, $s(v)$. Note that the elementary block's behavior illustrated in Fig. 4(c) is exaggerated with a large value of C so that it can clearly show how the slipping dynamics provides a smooth transition from the pre-sliding to the gross-sliding regime. From the parallel structure, we obtain the total estimation of friction force simply by summing the resulting force from each block as shown in the formula as F_f where N represents the number of elementary blocks. Owing to its straightforward structure, the GMS model provides a relatively simple implementation for the purpose of real-time control, and it has proven to describe many frictional phenomena experimentally discovered. In this paper, therefore, we work with the GMS model to estimate and compensate the friction of the experimental setup introduced in Section 2.1.

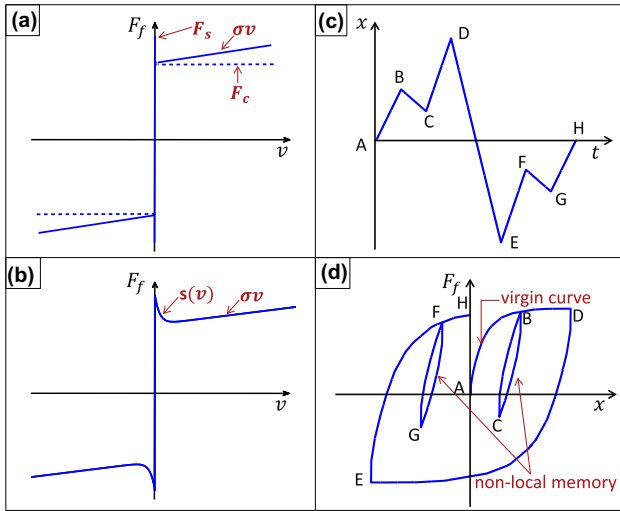


Fig. 3. Representative frictional behavior and simulation of a motion trajectory and resulting frictional force. (a) Stiction F_s , Coulomb friction F_c , and viscous friction σv . (b) Friction model of the Stribeck curve in the sliding regime with function $s(v)$ and σv . (c) Bidirectional position trajectory with small motions and turnarounds. (d) Hysteretic elastic behavior of the pre-sliding friction with non-local memory characteristics associated with motion (c).

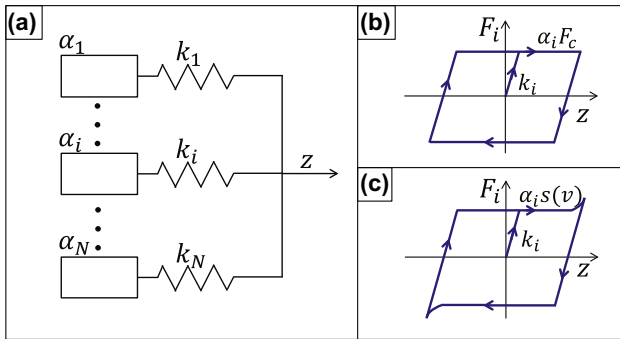


Fig. 4. Schematic of the Maxwell-Slip model with stiffnesses k_i , normalized saturation limits α_i , and friction forces F_i . (a) A series of Maxwell-Slip elementary blocks in parallel where z is the displacement in the pre-sliding regime. (b) Behavior of a Maxwell-Slip block with the Coulomb slip law. (c) Behavior of a Generalized Maxwell-Slip block with the slipping dynamics.

4. Friction parameter ID via resonance

The GMS model, despite the advantages of high fidelity and straightforward implementation, requires nontrivial effort to identify the many associated parameters. Here we discuss a frequency domain approach for parameter identification using frictional resonances. We call this the *Dahl resonance method* in this paper.

The parameters of the GMS friction model can be identified directly in the frequency domain since the nonlinear elastic effects of the pre-sliding friction are well-observed in the frequency responses. As illustrated in Fig. 6, the open-loop plant of the servomotor setup has varying frequency responses depending on the current command input amplitude. The response has a smaller DC gain and a resonance bump moves toward high frequency as the applied input amplitude decreases. This is consistent with the GMS nonlinear spring with increasing stiffness for smaller inputs. The resonant bump, which we call *Dahl resonance* named after Dahl who first investigated the pre-sliding friction [16,17], can be considered as the resonance of the nonlinear elastic behavior of the pre-sliding friction against the moving inertia. That is, the nonlinearity in the frequency responses is due to the hysteretic

elasticity of the pre-sliding friction, and the information on its changing stiffness is contained in the Dahl resonances. This is the principle of the approach of the frequency domain parameterization method developed in this paper. The Dahl resonance method provides a straightforward way to identify friction model parameters in the frequency domain, comparable to the conventional time domain method of hysteretic profile fitting [19–21].

4.1. Measurements of Dahl resonance

In order to identify the correct parameters for the friction model, we need a proper configuration for Dahl resonance frequency response measurements. What is usually conducted, as shown in Fig. 5(a), is to directly excite the plant with a sinusoidal input and measure both the excitation and the response signals while sweeping a frequency range of interest. A frequency response is then obtained by computing the magnitude ratio and phase shift of these signals at each frequency. This simple configuration, however, often causes drift at the output for inertia-dominated systems such as servomotors since any DC offset at the input is integrated. It is this drift that is problematic when it comes to measuring correct frequency responses in the pre-sliding regime because the drift keeps pushing the motor away from pre-sliding.

While it may be possible to limit the drift issue using white or colored noise as the excitation signal, a position controller is generally implemented in order to control the output of the plant so as not to drift away. By closing a control loop, however, it might cause bias in the system identification, especially when the system is disturbed [22]. To address this, the closed loop configuration depicted in Fig. 5(b) is utilized herein. In this case, we use a low-bandwidth (≈ 1 Hz) controller so as to both eliminate the low frequency drift and reduce the possible bias. The drift is eliminated in the loop due to an infinite loop gain at zero frequency, even with a low-bandwidth controller. The possible bias is kept small since the control law is ignorable for higher frequencies than the bandwidth, which is set to approximately 1 Hz in the measurement configuration.

The low-bandwidth closed-loop configuration additionally helps to not affect the current command by large control effort. By connecting the excitation signal of a dynamic signal analyzer as a disturbance to the plant, therefore, we can control the amplitude of the applied current command in the frequency range of interest above 1 Hz. Also note in Fig. 5(b) that a negative sign is required to have a negative feedback for the system stability.

4.2. Dahl resonance ID method

The frequency responses obtained in the low-bandwidth closed-loop configuration are shown in Fig. 6. Here we observe

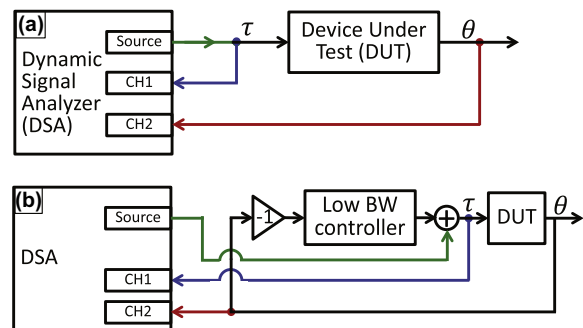


Fig. 5. Schematic configurations to obtain frequency responses of the servo system. (a) Open-loop setup allowing drift. (b) Low-bandwidth closed-loop configuration preventing low frequency drift.

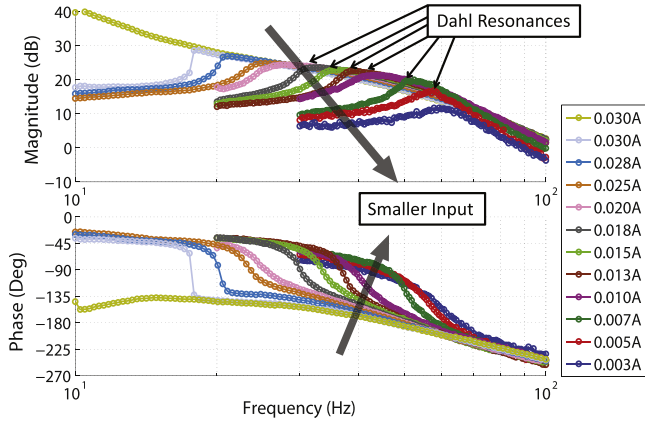


Fig. 6. Input-amplitude-dependent frequency responses of the open-loop plant with Dahl resonances. Curves are parameterized with the amplitude of the drive current output of the motor driver, which corresponds to motor torque. Magnitude is in units of [A/deg].

the Dahl resonances, which correspond to varying input amplitude levels of the current command, A_{cmd} . From this Dahl resonant frequency, f_{D_i} , we can identify the stiffness, k_i and the saturation limit, α_i of each of the GMS blocks. The stiffness information is directly obtained from the Dahl resonant frequencies using the mass moment of inertia of the rotor J and proper scaling for the unit conversions. The inertia of the motor provided by manufacturer is

$$J = 1.96 \times 10^{-5} \text{ [kg m}^2\text{]}.$$

Note that the inertia J can be also obtained from the measured frequency responses. The response curve for the relatively large input amplitude of 0.030 A shows an approximately straight magnitude plot with a -40 dB/decade slope, indicating a pure mass response in the sliding regime. By fitting this measured curve with an inertia model of

$$\frac{1}{J_{exp} s^2} \times K_t \times \frac{180}{\pi} \bigg|_{s=j\omega},$$

we can experimentally estimate the value of the inertia. Note that the measured responses are in the unit of [A/deg], so the inertia model is scaled by the factor of $\frac{180K_t}{\pi}$ to convert it to [Nm/rad] and calculate the mass moment of inertia in the unit of [kg m²]. The torque constant K_t is a given value of 0.2 [Nm/A], and it relates the phase currents of our three phase motor to the torque. Fig. 7 shows the measured responses together with the simulation curve of experimentally-obtained inertia $J_{exp} = 2.05 \times 10^{-5} \text{ [kg m}^2\text{]}$, which is closely equivalent to the value provided by the manufacturer. For the rest of the calculations, we use the nominal value of $J = 1.96 \times 10^{-5} \text{ [kg m}^2\text{]}$.

We then calculate the effective torsional stiffnesses by using the inertia J and the Dahl resonant frequencies f_{D_i} as

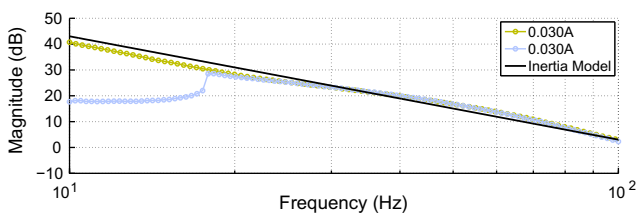


Fig. 7. Measured frequency responses and fitted simulation curve of an inertia model with $J_{exp} = 2.05 \times 10^{-5} \text{ [kg m}^2\text{]}$. Magnitude is in dB units of [A/deg].

$$K_{tors_i} = (2\pi f_{D_i})^2 \times J \text{ [Nm/rad]}$$

$$K_{eff_i} = \frac{K_{tors_i}}{K_t} \times \frac{\pi}{180} \text{ [A/deg]}.$$

The stiffness of the pre-sliding friction changes according to different input levels, and the effective torsional stiffness K_{tors_i} is defined as the representative stiffness that the system experiences at each input level. K_{eff_i} is a scaled value to match the unit to our measurements. These definitions are illustrated in Fig. 8.

The accumulated stiffnesses of GMS blocks in the ratio of the current to the angle are then obtained via

$$K_1 = K_{eff_1} \text{ [A/deg]},$$

$$K_{i+1} = \frac{A_{i+1} - A_i}{\theta_{i+1} - \theta_i} \text{ [A/deg]},$$

where $i = 1, 2, \dots, N-1$ is the number of the GMS blocks engaged in the slipping phase. We can now calculate the N stiffnesses k_i 's by taking the difference of the adjacent accumulated stiffnesses K_i 's as

$$k_i = K_i - K_{i+1} \text{ [A/deg]}$$

$$k_N = K_N \text{ [A/deg]}.$$

The saturation limits α_i 's are then obtained using the values of the calculated stiffness and the break points as

$$\alpha_j = k_j \times \theta_j \text{ [A]},$$

where $j = 1, 2, \dots, N$ is the index of each GMS block. To fit our experiments, we choose $N = 8$. The relevant variables and the identified parameters for the eight elementary blocks are organized in Table 1. We utilize these values for the friction compensation discussed in Section 5.

The measured frequency responses in Figs. 6 and 7 can also give an estimate of the static friction torque. As shown in the figures, two responses are measured for the current input amplitude of 0.03 A. The first measurement shows notable jumps at around 17 Hz in both the magnitude and phase plots, implying the transition from the pre-sliding regime to the sliding regime with increasing frequency. The second measurement conducted immediately after the first one shows a different response as if the system is in the sliding regime from the lowest frequencies. This transitional behavior of consecutive frequency responses implies that the current value of 0.03 A is on the verge of the breakaway point, and so the static friction has a very close value of $6 \times 10^{-3} \text{ Nm}$ for the servomotor setup with the torque constant of 0.2 Nm/A.

Note that Fig. 6 shows an interesting behavior at low frequencies in both the pre-sliding and sliding regimes. For input levels within the pre-sliding regime, the responses show phase offsets of approximately -25° as apposed to near zero for a standard mass-spring system. This is believed to be due to the nonlinear hysteretic behavior of the pre-sliding friction. That is, the spring-like behavior of the friction dominates over the inertia at low frequencies, but since the

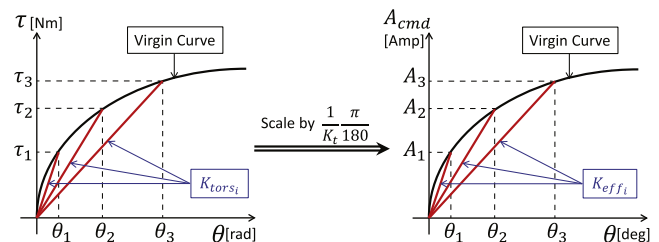


Fig. 8. Schematic describing the effective stiffnesses of the pre-sliding friction K_{tors_i} and K_{eff_i} with the scaling factors due to unit conversion. The left figure relates torque τ to θ in radians. The right figure relates current command amplitude A_{cmd} to θ in degrees.

Table 1

Relevant variables and identified parameters for the Dahl resonance method.

i	A_{cmd} [A]	f_{D_i} [Hz]	k_i [A/deg]	α_i [A]
1	0.003	60.74	0.0508	0.0306
2	0.005	57.85	0.0693	0.0767
3	0.007	52.49	0.0603	0.1134
4	0.010	42.69	0.0030	0.0123
5	0.013	38.95	0.0266	0.1686
6	0.015	35.33	0.0059	0.0525
7	0.018	31.53	0.0164	0.2196
8	0.020	27.68	0.0169	0.3264

frictional elasticity is not linear and has hysteresis, it causes a delay in the displacement following the input force. Similar behavior is also observed in the sliding regime response to the input level of 0.030 A. The Coulomb model for the sliding friction acts as a saturation block, adding a nonlinear damping to the system. This can cause input-level-dependent delay in the displacement at low frequencies where we expect a low-frequency phase of -90° for a standard mass-damper system. The delay becomes larger with increasing input level, and eventually for a sufficiently large input, the damping effect becomes negligible and the response would show a pure mass behavior.

To clarify these issues, we conducted a simulation using our friction model. We used the model to generate simulated frequency response curves which are in general agreement with the measured data, as shown in Fig. 9. Note that for an input level (0.001 A) small enough to consider the frictional stiffness to be linear (i.e. no GMS model blocks are slipping), the phase starts almost at 0° , and for a large input (0.300 A) the phase begins at almost -180° at low frequencies, showing the behavior of a nearly pure mass.

Another thing to note in the measured responses in Fig. 6 is a large phase delay. It seems that there are calculation delays and IO delays at certain points in the measurement loop, most likely in the motor driver. We need to consider this phase delay for the controller design described in Section 5.1. However, it is not problematic for the Dahl resonance ID method since we are interested solely in the Dahl resonant peaks to determine the necessary parameters for the friction model.

In summary, to identify the parameters via Dahl resonances, we need to:

- (1) Measure Dahl resonances as the highest magnitude points in the frequency responses where each resonance curve corresponds to varying input levels of the drive current command to the servo.
- (2) Extract the effective stiffness information K_{eff_i} using the measured Dahl resonant frequencies f_{D_i} and the system inertia J .
- (3) Calculate the accumulated stiffnesses K_i of the Generalized Maxwell-Slip (GMS) blocks using values of input levels and break points.
- (4) Obtain the stiffness k_i and the saturation limit α_i of each GMS block through the calculations illustrated previously.

This Dahl resonance method is naturally compatible with frequency-domain measurements for loop shaping control techniques in that it does not require additional position-controlled experiments in the time domain as long as only the pre-sliding friction is considered. To create the sliding function curve requires additional experiments at various steady velocities. In addition, the Dahl resonance method can be readily used because such resonances in the frequency domain are observed in a number of systems including servomotors [23], linear stages [23,24], hard-disk drives [25–28], fast tool servos [29,30], and lithographic machines [31].

5. Friction compensation and experimental results

There have been various methods introduced in the literature to address the effects of the friction in control systems, which include both model-based and non-model-based compensation techniques [24,32–34]. A simple model-based feedforward compensation is used in this paper so as to effectively observe the performance of the friction model identified by the Dahl resonance method discussed earlier. The fidelity of the identified friction model and the control system performance are tested with the experimental setup, and the results are presented in both the frequency and the time domains.

5.1. Model-based feedforward control

The identified GMS friction model based on Dahl resonances is used in the control algorithm shown in Fig. 10 to compensate the frictional effects of the experimental setup. In the figure, the right-hand dashed box represents a hardware plant including the real friction which we try to estimate with the model. The left-hand box indicates the overall control algorithm containing a linear position controller and a model-based feedforward friction compensator.

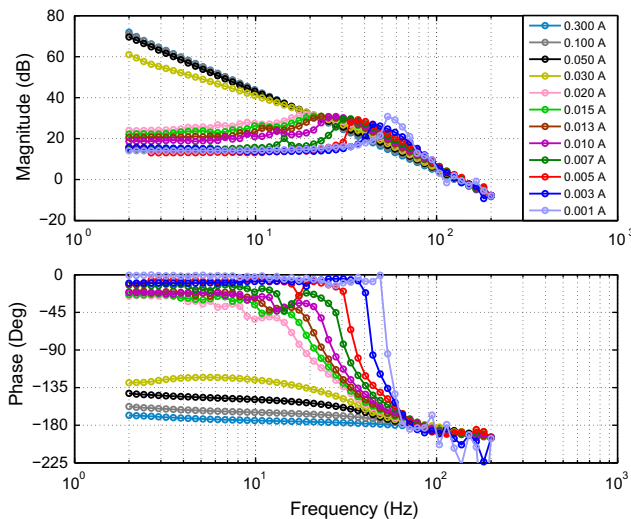


Fig. 9. Simulated GMS model frequency responses of the servo plant. Note similar responses to the measured data as shown in Fig. 6.

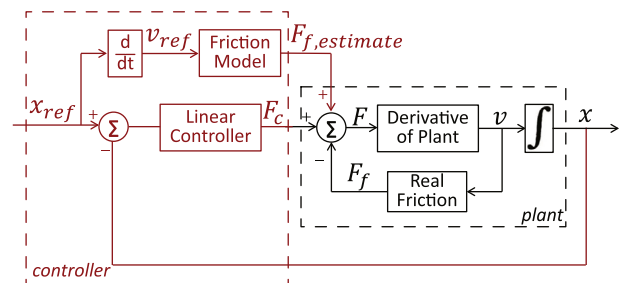


Fig. 10. Control algorithm schematic with a linear controller and a model-based feedforward compensator: real hardware plant (right-hand) and software controller in NI PXI chassis (left-hand). Trajectory x_{ref} and velocity v_{ref} are generated within the controller.

The linear position controller for the setup is designed by loop-shaping to have a crossover frequency of 50 Hz and a phase margin of 30° . The crossover frequency must be chosen carefully for a nonlinear system with such input-amplitude-dependent frequency responses. This is because the Dahl resonances move depending on the input amplitudes. We choose the crossover so as to minimize the Dahl resonance excitation as well as not to excite higher-frequency potential flexible modes of the system. We need to also consider the large excess phase delay in our choice of the crossover to guarantee a positive phase margin. The feedforward compensator utilizes the identified GMS model to estimate the real friction using the velocity command input and thereby reduce the effects of friction in the system.

5.2. Performance in the frequency domain

The nonlinear hysteretic elasticity of the pre-sliding friction is the cause of the input-dependent frequency responses. This effect deteriorates the closed-loop system bandwidth as experimentally confirmed with the closed-loop frequency responses in Fig. 11(a). The designed bandwidth is not achieved, with a significant droop in each response for position commands in the pre-sliding regime, approximately up to the amplitude of 0.4° for our servomotor setup. This undesired outcome is resolved to a significant extent in the feedforward compensated system as shown in Fig. 11(b). The droop in the closed-loop frequency responses is notably reduced with the friction compensation, and the responses of the compensated system become less dependent on the input level. These results in the frequency domain show that the Dahl resonance identification method provides reasonable values for the parameters of the friction model.

5.3. Performance in the time domain

The system performance improvements are also observed in the time domain via tracking performance. When a control system is commanded to follow a position trajectory, the servo system must travel the pre-sliding regime where the nonlinear frictional stiffness becomes dominant. Without a proper compensation, therefore, this dominant nonlinear effect would not be overcome by any linear controller, resulting in poor tracking, especially noticeable in high-precision machines. We ran tracking experiments to observe the limitation of a linear controller and the improvement

by the friction compensator. In the experiments, we commanded sinusoidal position references with the frequency of 0.5 Hz and the amplitude of 7° peak for the sliding regime and 0.12° for the pre-sliding regime. The results are compared in Fig. 12. The tracking performance is significantly improved both in the pre-sliding and the gross-sliding regimes in both RMS error reduction and peak error reduction, as presented in Table 2. The peak error reduction is more important in the sliding regime experiments rather than the RMS error since the largest tracking errors occur in the velocity reversals and the friction compensation aims to reduce those peak errors by eliminating the pre-sliding frictional effects.

There still remains some response distortion in the frequency domain and position error in the time domain shown in Figs. 11 and 12 respectively. This might be because the linear controller implemented may not be able to provide a perfect trajectory following, thereby causing discrepancy between the position command and the actual output while the model-based friction compensator uses the derivative of the position reference instead of the real velocity. Within these limitations, however, the identified friction model shows significant improvements, indicating the high fidelity of the friction model identified by the Dahl resonance method.

6. Conclusions and future work

This paper studies aspects of friction both in the sliding and pre-sliding regimes, develops an identification method in the frequency domain, and discusses a friction compensation technique using the GMS friction model. The model-based compensation shows promising results when the model parameters are identified with the Dahl resonance method performed in the frequency domain. This frequency domain approach significantly improves the performance of the friction-compensated system, proving that it can be readily used with the equivalent benefits of the conventional method of the hysteresis profile fitting. The performance results experimentally compared in both the frequency and the time domains for both friction regimes help understand how friction affects a control system and how we can compensate it to achieve higher accuracy for high-precision motion control systems.

As future work, we suggest that the experimentally identified friction model can be incorporated with a linear finite element method (FEM) model of any structural modes to yield a full system analysis. Connecting the nonlinear friction model block to the FEM

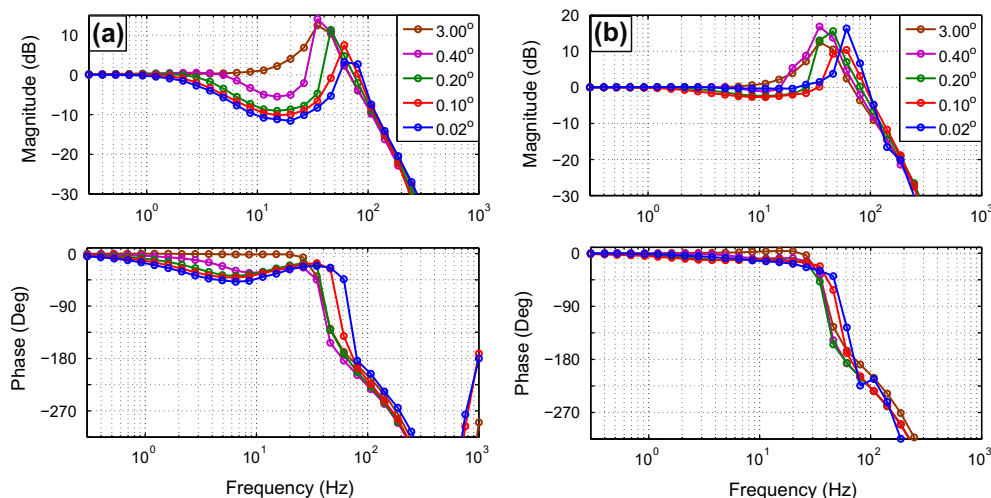


Fig. 11. Measured closed-loop frequency responses for varying amplitude of angular displacement. (a) Friction-uncompensated system. (b) Friction feedforward compensated system.

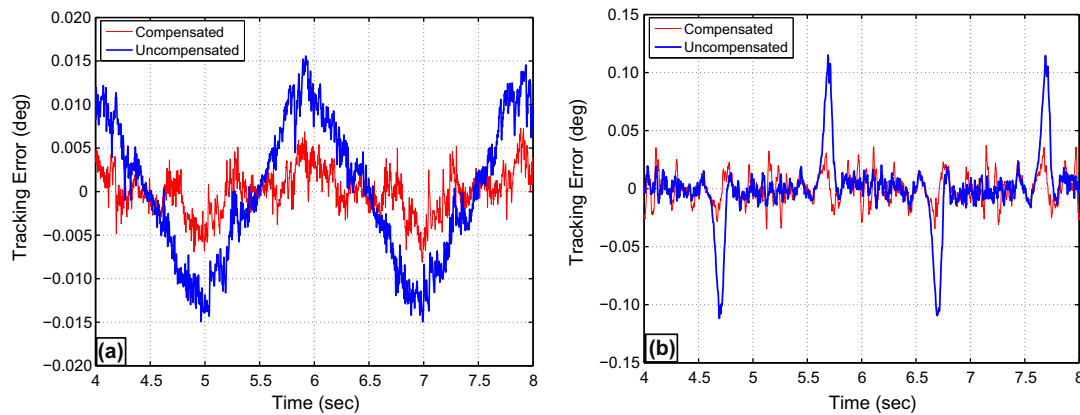


Fig. 12. Tracking performance comparison in position error between friction-uncompensated system (blue) and feedforward compensated system (red). (a) In the pre-sliding regime. (b) In the gross-sliding regime. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Tracking error reduction of the friction compensated system.

	RMS error reduction (%)	Peak error reduction (%)
Pre-sliding regime	68.0	53.4
Sliding regime	56.8	73.2

and observing the behavior of friction model prediction in both simulations and experiments can be considered as a future work. In addition, it would be also an interesting research effort to predict the frictional stiffness of a system from first principles, using an accurate model for mechanical components in contact, such as bearings, thereby allowing the use of such models in the machine design phase.

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